

Project Plan: Sentiment Analysis of Modern and Traditional Architecture in China

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1. Introduction

This project aims to construct a robust and scalable data pipeline to collect, process, and analyze Instagram posts for assessing public sentiment regarding modern and traditional architectural styles in China. By integrating natural language processing (NLP) for sentiment analysis and computer vision for image analysis, this study seeks to provide comprehensive insights into the perception of architectural aesthetics from social media platforms.

2. Project Workflow

Phase 1: Data Collection

Step 1.1: Instagram Data Acquisition

1. Configure Instagram Graph API

- Secure API credentials and permissions.
- Establish API requests to extract public posts containing targeted hashtags associated with modern and traditional Chinese architecture.

2. Data Extraction

- Collect a dataset of at least 3,000 posts, including:
 - Captions
 - Image URLs (if accessible)
 - Geolocation metadata
 - Engagement metrics (e.g., views, likes, shares, comments)

Phase 2: Data Processing & Preprocessing

Step 2.1: Text Data Preprocessing

1. Data Cleaning & Normalization

- Remove extraneous elements such as URLs, special characters, and non-relevant text.
- Apply tokenization and lemmatization.

Step 2.2: Sentiment Analysis

1. Model Selection & Application

- Utilize lightweight transformer-based models (e.g., BERT, RoBERTa, or fine-tuned Chinese sentiment analysis models).

- If hardware permits, can also incorporate open source LLMs such as Llama 3.2 and DeepSeek -V3
- If funding permits, can use more advanced closed-source model APIs such as GPT-4o and Google Gemini.

Phase 3: Documentation & Research Contribution

Step 3.1: Methodology Documentation

1. Detailed reporting on data collection, processing, and analytical methodologies.
2. Incorporation of visualizations and key findings (if time permits).

Step 3.2: Research Paper Preparation

1. Summarization of findings in an academic format.
2. Collaboration on manuscript preparation for a research publication.

Phase 4: Project Completion & Deliverables

Step 4.1: Code Implementation & Analysis

1. Automate data retrieval and processing pipelines.
2. Compute engagement metrics and execute sentiment analysis from comments and derive insights.
3. Generate visual representations using Matplotlib and Seaborn (if time permits).

Step 4.2: Final Deliverables

1. Curated dataset structured for further academic exploration.
2. Comprehensive sentiment analysis results with visualized trends.
3. Final research report summarizing methodologies and findings.

3. Timeline & Milestones (20 – 25 Hours Total)

Step 1: Data Collection (4 – 5 hours)

Step 2: Preprocessing & Analysis (~15 hours or more depending on the amount of data, the model used and the hardware – more accurate results require more time)

Step 3: Documentation & Report Writing (~ 2 hours)

4. Tools & Technologies

Programming language: Python

API: Instagram Graph API, Python |

NPP: BERT, RoBERTa, DeepSeek-V3 and Llama-3.2 (if hardware permits), OpenAI GPT models (if funding permits)

Image Analysis: OpenCV, PyTorch

Data Visualization: Matplotlib, Seaborn

5. Summary

This project is designed to systematically evaluate public sentiment towards modern and traditional Chinese architecture through multimodal analysis of social media content. By leveraging text and image processing techniques, the study can offer actionable insights for academia, urban planners, and architectural designers. The final deliverables will contribute to a research publication.

6. Future Works

If there are future fundings, Little Red Book data can be collected and analyzed in a similar way as the Instagram data. Comparisons can be made across these two different platforms to gain more insights, for example, on how audience from different platforms share different or similar sentiments on Chinese architecture.